Completed Research:

Using AI for Detecting Clinical Deterioration: Insights and Responses from the Care Team

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BACKGROUND AND AIM

Background: In 2021, an AI-based clinical deterioration prediction model was implemented at a major academic medical center (Stanford Health Care), reducing absolute risk of care escalations among inpatients by 10.4% over two years.

Aim: To evaluate the experiences of nurses and physicians with the AI-enabled clinical deterioration pathway, including the prediction tool and subsequent workflows

METHODS

Study Design: A comprehensive survey was administered to evaluate the care team's perspectives on the AI-enabled workflow.

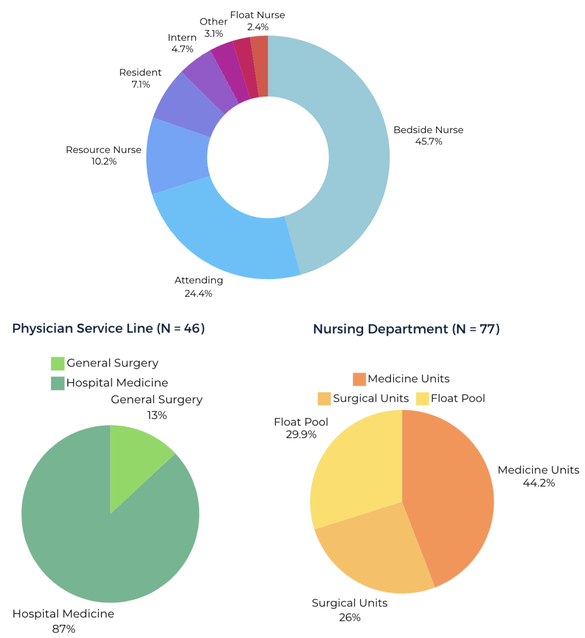
Analysis Techniques: Statistical analysis of survey results (one sample T-test, Kruskal-Wallis Test) with free-text responses thematically analyzed using Stanford's GPT-4 secure instance

Setting and Population: The study took place at Stanford Health Care with a survey cohort of 127 physicians, residents, interns, bedside nurses, resource nurses, and float nurses from general surgery and hospital medicine units.

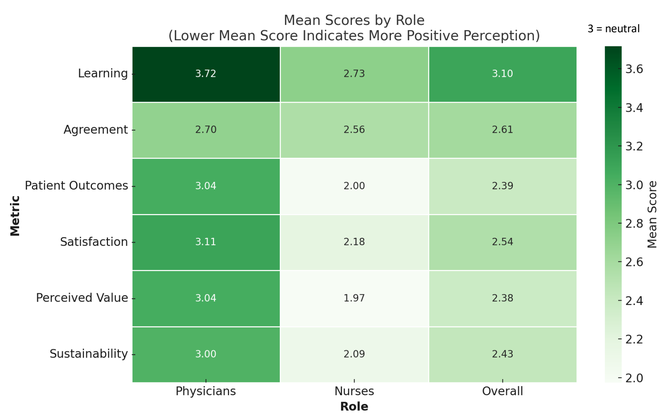
Frameworks and Statistical Tools: Evaluation through SEIPS 2.0 and RE-AIM frameworks

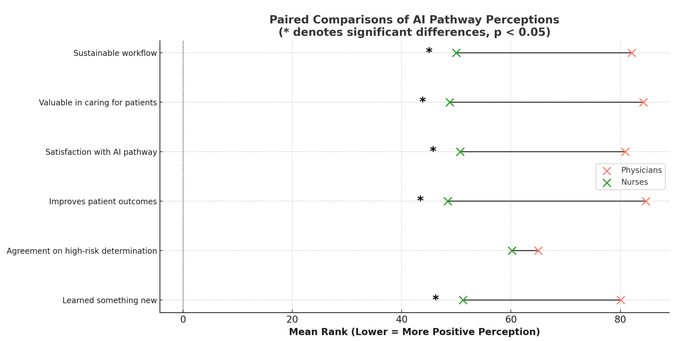
RESULTS: IMPACT ON EXPERIENCE

Demographic Summary

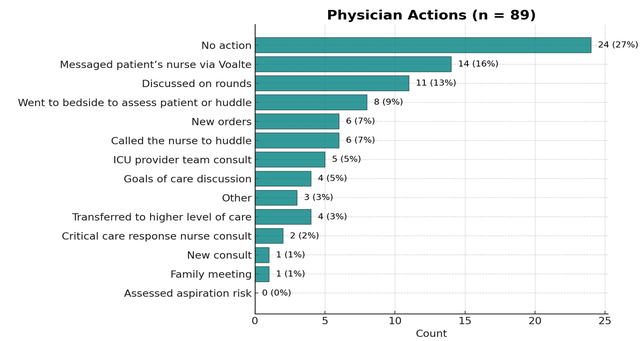


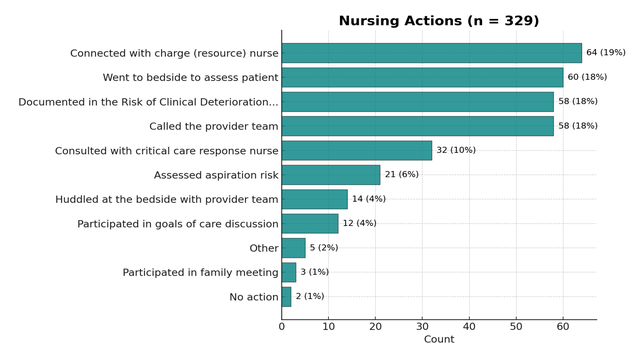
Survey Results Summary



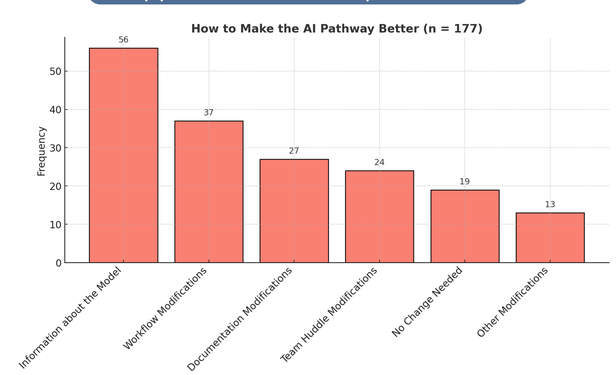


Implementation: Actions





Opportunities for Improvement



Free Text Summary Analysis was conducted using Stanford's Secure ChatGPT.

- **Model Information**: Users emphasized the need for precise, accurate, and adaptable model functionality, especially for chronic patient cases.
- **Workflow Modifications:** Feedback suggested clearer procedures after alerts, regular training, reduced alert frequency, and minimized workflow disruptions.
- **Documentation Modifications:** Users requested more relevant, precise, and streamlined documentation.
- **Team Huddle Modifications:** Recommended clearer roles, better communication, and regular meetings to strengthen team discussions on the AI pathway.
- **No Change Needed:** Pathway deemed effective in its current form.
- **Other Modifications:** Suggested improvements in notifications, UI, support tools, tailored training, and adaptation for various patient groups.

KEY LEARNINGS

High Nurse Engagement: Nurses reported higher satisfaction and engagement than physicians, suggesting a need to tailor support and training to each role's unique requirements.

Opportunities for Improvement: Feedback emphasizes the need for ongoing education on AI model functionality, adjusted workflows for reduced alert fatigue, and enhanced team huddle communication.

Pathway to Enhanced Adoption: Overall satisfaction was positive, with specific suggestions pointing to a need for more integration support to improve adoption and engagement.

NEXT STEPS

Qualitative Insights: Conducted interviews with 36 care team members (12 physicians, 24 nurses) to deepen understanding of user feedback and barriers to adoption. Qualitative analysis will provide further insights into improving AI integration.

Refinement of Workflow: Based on survey feedback, future efforts will focus on refining alert processes, enhancing documentation protocols, and optimizing team huddle structures for clearer communication.